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Career Exploration 5: Artificial Intelligence and the Future of the Workforce

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ARTIFICIAL INTELLIGENCE AND THE FUTURE OF THE WORKFORCE



What is Artificial Intelligence?



Risky Business



A Look Into Your Future



Dealing with Uncertainty

What is Artificial Intelligence?

GOALS

RESOURCES

- Determine which jobs are at low, medium, and high risk of being replaced by artificial intelligence.
- Explain how people can prepare themselves for advances in artificial intelligence in the workforce.

GOALS

RESOURCES

- [Information Credibility](#)
- [Career Exploration 5: Artificial Intelligence and the Future of the Workforce \(Downloadable PDF\)](#)



“Artificial Intelligence (AI) is a branch of computer science that focuses on creating intelligent machines that can perform tasks that typically require human intelligence, such as understanding natural language, recognizing images, and making decisions. AI can be applied to various sectors, such as medicine, finance, and entertainment, to enable new possibilities and efficiencies. AI is important because it can automate and enhance tasks that require human capabilities, such as understanding, reasoning, planning, communication, and perception. AI can also perform frequent, high-volume, computerized tasks with fatigue or breaks.”

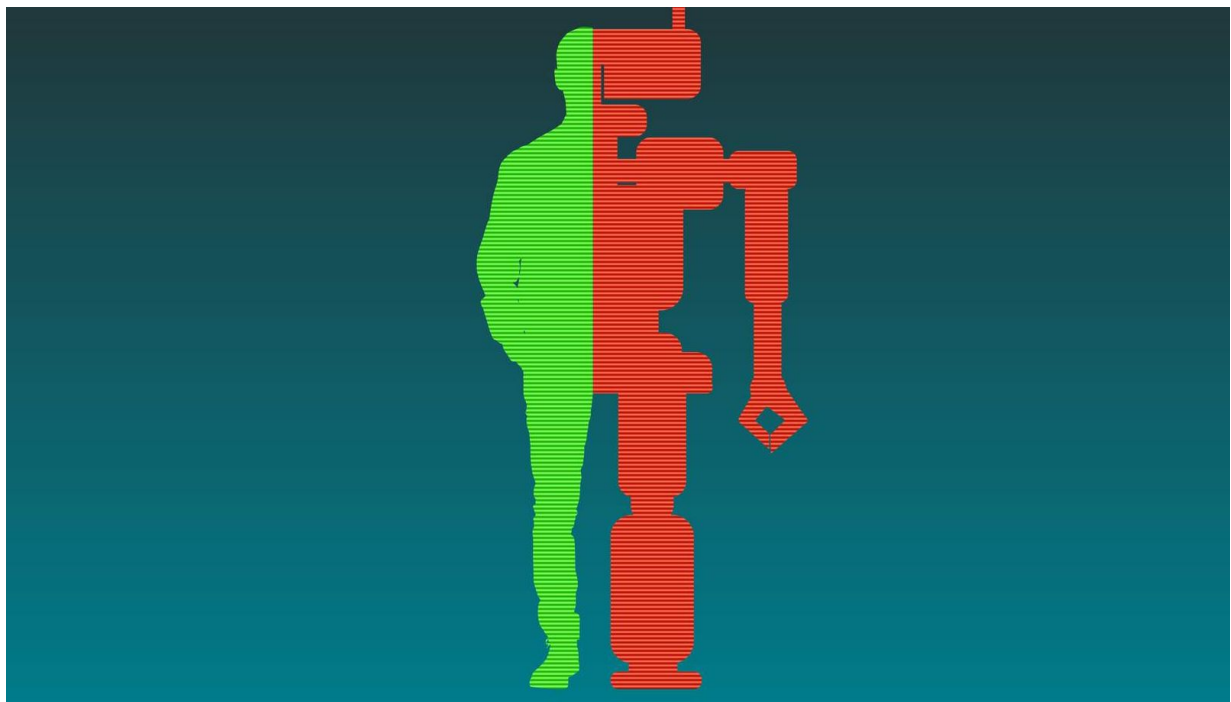
- GPT-4, a modern AI on the topic of AI

When you think about the future of the workforce, what do you envision? Do you see offices full of workers or people working remotely? Are factories, restaurants, and grocery stores operated by people or automated robots? Have some jobs been eliminated due to changes in technology? While it is hard to predict what the future of the workforce will look like, advances in AI ensure that change is inevitable.

In this activity, you will learn about job categories that are at risk of being changed or eliminated due to AI. You will also determine whether the careers you researched in Career Exploration 1 are categorized as high, medium, or low risk.

Risky Business

Currently, AI is skilled at replicating human abilities in critical thinking, data analysis, writing, science, and mathematics. As a result, jobs that rely on those skills are more likely to be replaced by AI in the future. According to a 2022 study conducted by the Pew Research Center, the jobs in the table below are at medium and high levels of risk of being eliminated due to the abilities of AI. While the list of jobs is not exhaustive, it highlights various industries that are expected to be impacted by AI.



Medium Risk

High Risk

Interior Designers	Web Developers
Chief Executives	Accountants and Bookkeepers
Sales Managers	Budget Analysts
Marketing Managers	Technical Writers
Credit Authorizers	Data Entry Specialists
Postsecondary Teachers	Judicial Law Clerks and Paralegals
Teaching Assistants	Architectural and Civil Drafters
School Psychologists	Proofreaders and Copywriters
Customer Service Representatives	Real Estate Title Examiners and Abstractors
Materials Engineers	Medical Transcriptionists

Turn and Talk



- Do any of the risk designations in the table above surprise you?
- What jobs would you add to the medium risk column of the table?
- What jobs would you add to the high risk column of the table?

Low-Risk Careers

Not all jobs are at the same level of risk from advances in AI. Jobs that are more physically intensive are thought to have the lowest level of risk in the future. In the table below, several low-risk jobs are listed.

Low Risk		
Nurses	Physical Therapists	Athletic Trainers
Orthotists and Prosthetists	Emergency Medical Technicians	Firefighters
Landscapers	Police Officers and Detectives	Dog Trainers
Surgeons	Fish and Game Wardens	Construction Workers

Turn and Talk



What other low risk jobs would you add to the table above? Explain your thinking.

A Look Into Your Future

In Career Exploration 1, you researched two careers that interested you. As you plan for postsecondary learning experiences, it is important to understand how AI may influence the future of these career pathways.

1

Conduct Internet research to determine how advances in AI may impact each career.

NOTE: As you conduct research, take into account the credibility of the website you are visiting. For help determining whether a website is reliable, review **Information Credibility**.

2

Categorize each career as either high, medium, or low risk of being eliminated or modified due to AI.

Reflection



Did your findings change your level of interest in either career path? Why or why not?

Dealing with Uncertainty

Even though experts predict that certain jobs may be eliminated in the future due to AI, nothing is certain. However, it is wise to prepare yourself for an uncertain future. To best prepare yourself for the future of AI technology experts suggest that you do the following.

- Become as familiar as you can with current technology trends.
 - Stay up to date by reading technology journals and blogs, listening to podcasts, finding a mentor in industry, and attending events.
- Use advancing technologies to amplify your skill sets.
 - Enroll in a training course or micro-credential class.
- Develop a growth mindset.
 - Prioritize flexibility and adaptability as core skills in your everyday actions.
- Learn how to verify information provided by AI technologies.

- Be skeptical when something sounds too good to be true.
- Consult credible sources to validate information.
- Learn how to interact with current AI models.
 - Practice using AI solutions such as ChatGPT.
- Learn to code.
 - *Python*® is a great language to start with!

As you make decisions about your future, continue to consider the impact that AI may have. While the world is at your fingertips, it is constantly evolving.



Did You Know?

Today's leaders in technology disagree on the impact AI will have on the future workforce. Mark Zuckerberg and Jeff Bezos share excitement over the helpful functions of AI and are not worried about job elimination. On the other hand, Elon Musk, Stephen Hawking, and Bill Gates share concerns about evolving intelligence systems that perform better than humans, surpass human intelligence, and eliminate jobs.

Turn and Talk



To what extent are you concerned about the impact of AI on the future workforce?